SM015
Mathematics 1
Semester I
Session 2022/2023
2 hours

SM015
Matematik 1
Semester I
Sesi 2022/2023
2 jam

KEMENTERIAN PENDIDIKAN

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **12** halaman bercetak.
*This question paper consists of **12** printed pages.*

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **Section A** and **Section B**.

Section A consists of **3** questions. **Section B** consists of **7** questions.

Answer **all** questions.

All answers must be written in the answer booklet provided. Use a new page for each question.

Marks for each question or section are shown in the bracket at the end of the question or section.

All steps must be shown clearly.

The use of non-programmable scientific calculator is permitted.

Numerical answers may be given in the form of π, e, surd, simplest fractions or up to three significant figures, where appropriate, unless stated otherwise in the question.

ARAHAN KEPADA CALON:

Kertas soalan ini mengandungi ***Bahagian*** **A** *dan* ***Bahagian*** **B**.

Bahagian **A** *mengandungi* **3** *soalan.* ***Bahagian*** **B** *mengandungi* **7** *soalan.*

Jawab ***semua*** *soalan.*

Semua jawapan hendaklah ditulis pada buku jawapan yang disediakan. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Markah yang diperuntukkan bagi setiap soalan atau bahagian soalan ditunjukkan dalam kurungan pada penghujung soalan atau bahagian soalan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas.

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

Jawapan berangka boleh diberi dalam bentuk π, e, surd, pecahan termudah atau sehingga tiga angka bererti, di mana-mana yang sesuai, kecuali jika dinyatakan dalam soalan.

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned} \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A \end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Quadratic equation $ax^2+bx+c=0$:
Persamaan kuadratik $ax^2+bx+c=0$:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Differentiation
Pembezaan

$f(x)$	$f'(x)$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x\tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x\cot x$

$$\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}$$

$$\frac{d^2y}{dx^2}=\frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Sphere ***Sfera***	$V=\frac{4}{3}\pi r^3$	$S=4\pi r^2$
Right circular cone ***Kon membulat tegak***	$V=\frac{1}{3}\pi r^2h$	$S=\pi r^2+\pi rh$
Right circular cylinder ***Silinder membulat tegak***	$V=\pi r^2h$	$S=2\pi r^2+2\pi rh$

SECTION A [25 *marks*]

BAHAGIAN **A** [25 *markah*]

This section consists of **3** questions. Answer **all** questions.

Bahagian ini mengandungi **3** *soalan. Jawab* ***semua*** *soalan.*

1 Solve

Selesaikan

(a) $\lim\limits_{x \to -3} \dfrac{x^3 + 27}{x + 3}$.

[4 *marks*]

[4 *markah*]

(b) $\lim\limits_{x \to 2} \dfrac{\sqrt{x^2 + 12} - 4}{x - 2}$.

[5 *marks*]

[5 *markah*]

2 By expressing $y = \dfrac{1}{(x-1)^2(x-3)}$ into partial fractions, find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. Give your answer in the simplest form.

Hence, evaluate $\dfrac{d^2y}{dx^2}$ when $x = -1$.

Dengan mengungkapkan $y = \dfrac{1}{(x-1)^2(x-3)}$ *dalam bentuk pecahan separa, cari* $\dfrac{dy}{dx}$ *dan* $\dfrac{d^2y}{dx^2}$. *Beri jawapan anda dalam bentuk teringkas.*

Seterusnya, nilaikan $\dfrac{d^2y}{dx^2}$ *apabila* $x = -1$.

[9 *marks*]

[9 *markah*]

3 Find the critical numbers for $f\left(x\right)=x^4-8x^3+18x^2-27$. Hence, determine the points of inflection.

Cari nombor-nombor genting bagi $f\left(x\right)=x^4-8x^3+18x^2-27$. *Seterusnya, tentukan titik-titik lengkuk balasnya.*

[7 *marks*]

[7 *markah*]

SECTION B [75 *marks*]

***BAHAGIAN* B** [75 *markah*]

This section consists of **7** questions. Answer **all** questions.

Bahagian ini mengandungi **7** *soalan. Jawab* ***semua*** *soalan.*

1 Given $z_1 = 6+3i$ and $z_2 = 5-2i$.

Diberi $z_1 = 6+3i$ *dan* $z_2 = 5-2i$.

(a) Verify $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$.

Tentusahkan $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$.

[3 *marks*]

[3 *markah*]

(b) Find $\text{Arg}(z_1)$.

Cari $\text{Arg}(z_1)$.

[2 *marks*]

[2 *markah*]

2 Solve

Selesaikan

(a) $x^{\ln x - 2} = e^3$.

[7 *marks*]

[7 *markah*]

(b) $|2x+3| > |x-5|$, $x \neq 5$.

[4 *marks*]

[4 *markah*]

3 Given

Diberi

$$f(x) = 4^{x+2}.$$

(a) Find the value of a when $f(a) = 1$.

Cari nilai a *apabila* $f(a) = 1$.

[2 *marks*]

[2 *markah*]

(b) Find f^{-1} and its domain.

Cari f^{-1} *dan domainnya.*

[4 *marks*]

[4 *markah*]

4 Given

Diberi

$$f(x) = e^{-2x} + 3;\ (g \circ f)(x) = \frac{1}{2}\ln\left(e^{-2x} + 2\right).$$

(a) Find domain and range of $f(x)$.

Cari domain dan julat bagi $f(x)$.

[3 *marks*]

[3 *markah*]

(b) Find $g(x)$.

Cari $g(x)$.

[5 *marks*]

[5 *markah*]

(c) Hence, find $g^{-1}(x)$. Sketch the graph of $g(x)$ and $g^{-1}(x)$ on the same axes.

Seterusnya, cari $g^{-1}(x)$. Lakarkan graf bagi $g(x)$ dan $g^{-1}(x)$ pada paksi yang sama.

[6 *marks*]

[6 *markah*]

5 (a) Evaluate

Nilaikan

$$\lim_{x \to -\infty} \frac{3x+2}{\sqrt{4x^2-3}}.$$

[4 *marks*]

[4 *markah*]

(b) Given

Diberi

$$f(x)=\begin{cases} ax+3, & x>5, \\ 8, & x=5, \\ x^2+bx-2a, & x<5. \end{cases}$$

Find a and b such that the function is continuous for all x.

Cari a dan b supaya fungsi selanjar untuk semua x.

[4 *marks*]

[4 *markah*]

(c) Given

Diberi

$$y = \frac{x+1}{(x-4)(x+2)}.$$

By finding the limit at appropriate points, obtain the horizontal and vertical asymptotes.

Dengan mencari had pada titik yang sesuai, dapatkan asimptot mencancang dan mengufuk.

[6 *marks*]

[6 *markah*]

6 (a) Given

Diberi

$$y^3 = e^{x-y}.$$

Show that

Tunjukkan bahawa

(i) $$\frac{dy}{dx} = \frac{y}{3+y}.$$

[4 *marks*]

[4 *markah*]

(ii) $$(3+y)^3 \frac{d^2y}{dx^3} - 3y = 0.$$

[4 *marks*]

[4 *markah*]

(b) Given the parametric equations for the curve

Diberi persamaan berparameter bagi suatu lengkung

$$x = 2 + \sqrt{\cos 2t} \text{ and } y = 1 - \sqrt{\sin 2t}.$$

Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. Give your answer in the simplest form.

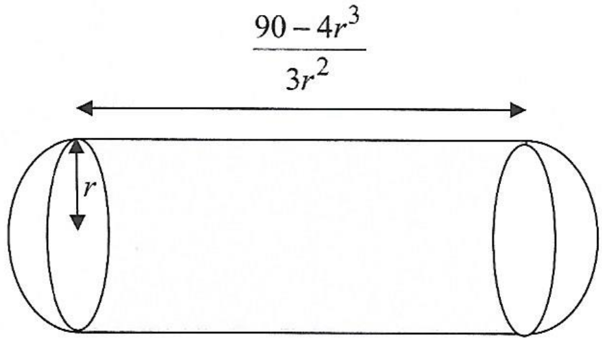
Cari $\dfrac{dy}{dx}$ *dan* $\dfrac{d^2y}{dx^2}$. *Beri jawapan anda dalam bentuk teringkas.*

[8 *marks*]

[8 *markah*]

7 The figure below shows a cylinder with both ends are two hemispheres with radius, r meter and length $\dfrac{90-4r^3}{3r^2}$ meter.

Rajah di bawah menunjukkan sebuah silinder dengan kedua-dua hujungnya ialah hemisfera berjejari, r *meter dan panjang* $\dfrac{90-4r^3}{3r^2}$ *meter.*



(a) Show that the total surface area is

Tunjukkan bahawa jumlah luas permukaan ialah

$$A = \frac{4}{3}\pi r^2 + \frac{60\pi}{r}.$$

[3 *marks*]

[3 *markah*]

(b) Hence, find the value of r when the total surface area is minimum.
Seterusnya, cari nilai r apabila jumlah luas permukaan adalah minimum.

[6 *marks*]
[6 *markah*]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT